

Congested Traffic Flow Observations and Theory

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Recent and some new results of observations of traffic flow dynamics in congested conditions are considered. On the basis of these results, hypotheses about properties of congested flow are proposed and discussed. In particular, it is proposed that the complexity of congested flow is linked to (a) the existence of three qualitatively different phases in traffic flow: free flow, synchronized flow, and wide traffic jams; (b) an occurrence of three qualitatively different types of phase transitions between these three phases of traffic flow; (c) the result that all these qualitatively different phase transitions are related to the same class of “first-order local phase transitions;” (d) complex dynamical behavior of synchronized flow; and (e) the existence of two qualitatively different kinds of “nucleation effects” in traffic flow: the nucleation effect responsible for the jam’s formation, which is linked to an avalanche increase in the amplitude of an initial local perturbation of traffic variables (vehicle speed, density, or both), and the one responsible for the phase transition from free flow to synchronized flow, which is linked to an avalanche decrease in the probability of passing in free traffic flow.

A common basis of almost every traffic flow theory is the hypothesis about the fundamental diagram (1–6). The fundamental diagram is a curve in the flow-density plane that relates a given vehicle density to the corresponding flow rate. At low vehicle density, the slope of the fundamental diagram is positive, and at high vehicle density, it is negative. It is often proposed that the fundamental diagram consists of two isolated curves separated by a gap (the problem of capacity drop): a curve with a positive slope for *free* flow (in the U.S. literature this case is often referred to as uncongested or undersaturated flow) and a curve with a negative slope for *congested* flow (note that in the frame of the hypothesis of the fundamental diagram the related average vehicle speeds in congested flow include at least a momentary stop together with periods of movement) (7).

Because of relatively low densities of vehicles in free flow, drivers on a multilane road are able to pass, whereas because of the higher density in congested flow, vehicles are almost unable to pass. Therefore, when all lanes of a highway correspond to the same route without on- and off-ramps or other bottlenecks, drivers move with nearly synchronized average speed on the different lanes of the highway. The possibility of such *synchronized* (collective) flow, which is related to a part of the fundamental diagram at higher density, has been theoretically predicted by Prigogine and Herman (1). Synchronized flow has been observed by Koshi et al. (8).

However, it is well known that within congestion, the flow-density data do not form a neat relationship but show a broad and complex spreading of measurement points; that is, the flow-density data cover a two-dimensional region on the flow-density plane (8). This *complexity* in traffic flow is usually interpreted as fluctuations, or as an instability, or else as traffic jam formation (3–8).

REASONS FOR COMPLEXITY IN TRAFFIC FLOW

Recently Kerner and Rehborn (9) showed that on the basis of experimental observations, in contrast to free flow, synchronized flow shows very complex dynamic behavior. In particular, even the multitude of homogeneous states of synchronized flow (i.e., states that are homogeneous spatially and stationary in time, often called “steady speed” states) cover a two-dimensional region in the flow-density plane: a given vehicle speed (a steady speed) in a homogeneous and stationary state of synchronized flow may be related to an infinite multitude of vehicle densities, and a given density may be related to an infinite multitude of different steady speeds. In other words, there are no fundamental diagrams that may describe the whole multitude of even homogeneous (steady) states of traffic flow.

Besides, it has been discovered (9,10) that in congested flow two qualitatively different phases of traffic must be distinguished—wide traffic jams and synchronized traffic flow—and that in free flow, two qualitatively different kinds of phase transitions must be distinguished—free flow $\Leftrightarrow$ jam and free flow $\Leftrightarrow$ synchronized flow. As a result, the complexity in traffic flow mentioned earlier is linked (a) to the complex behavior of synchronized flow and (b) to the existence of qualitatively different local first-order phase transitions, which can spontaneously occur among wide jams, synchronized flow, and free flow. (Note that the term “jam” designates “moving jam” throughout the paper.)

Thus, there are three qualitatively different phases in traffic: free flow, wide jams, and synchronized flow. Respectively, there are three different types of local first-order phase transitions: free flow $\Leftrightarrow$ jam, free flow $\Leftrightarrow$ synchronized flow, and synchronized flow $\Leftrightarrow$ jam(s). All of these phase transitions are accompanied by similar-looking breakdown and hysteresis effects because these phase transitions are related to the same class of first-order phase transitions (10). However, it is necessary to distinguish between them because the nonlinear features of these phase transitions are qualitatively different (10,11). This differentiation determines one of the main difficulties in the understanding of the observations and of traffic flow theory. In particular, the well-known stop-and-go phenomenon in traffic flow (8), as Kerner recently discovered (11), can be explained on the basis of a differentiation among the just-mentioned phase transitions in traffic flow.

Note that in contrast to synchronized flow, in which the vehicle speed is relatively lower but the flow rate can be nearly as high as the flow rate in free flow, inside a wide jam both the vehicle speed and the flow rate are very low, sometimes zero. Recall that a wide jam is a traffic jam whose width (i.e., the longitudinal distance between the jam fronts) is considerably larger than the widths of both upstream and downstream jam fronts. The jam front is a region where the flow rate, the density, and the average vehicle speed sharply change spatially. Note that the term “jam front” is used instead of “shock wave” because the latter term in traffic flow theory is usually associated

with the classical Lighthill-Whitham theory of shock waves (2). However, as Kerner et al. have recently shown (12), the Lighthill-Whitham theory cannot explain the properties of experimentally observed (13) traffic jams.

It must be noted that corresponding to general properties of first-order phase transitions in physical, chemical, and biological systems, the occurrence of a first-order phase transition in traffic flow (10) is linked to the “nucleation effect.” The nucleation effect in traffic flow is the avalanche growth of the amplitude of a local perturbation of one of the traffic flow characteristics. This avalanche growth will be realized only if the amplitude of the local perturbation exceeds some critical amplitude. A state of flow where the nucleation effect can occur is called a metastable state, which is stable with respect to any infinitesimal perturbations but in which critical perturbations are growing, that is, the perturbations whose amplitude exceeds the critical amplitude. There is a threshold of the phase transition where the critical amplitude is maximal and the related probability of the occurrence of this phase transition is minimal. Above the threshold, the greater the distance is from the threshold, the lower the critical amplitude and the higher the related probability of the occurrence of the phase transition.

In this paper, recent (9–11,13–16) and some new results of experimental observations of congested flow will be considered. On the basis of the results of these observations, hypotheses about properties of congested flow will be proposed and explained.

RESULTS OF EXPERIMENTAL OBSERVATIONS

Characteristic Line for Downstream Front of Wide Jams

Kerner and Rehborn (13) found that when “control” parameters of traffic (weather, other road conditions, number of lanes, etc.) are given, there are some characteristic (i.e., unique, reproducible, and predictable) parameters of wide jams that do not depend on initial conditions (i.e., on initial spatial distributions of the density and the average vehicle speed on a highway) or on fluctuations in traffic flow. In particular, if free flow is formed in the outflow of the jam, the jam has the following properties:

1. The downstream fronts of different wide jams possess nearly the same stationary moving structure. Thus, the mean values of the parameters of the downstream fronts (velocity, flow rate out of a wide jam, vehicle speed and density in outflow from the jam, vehicle density inside the jam) are nearly the same for different wide jams. These parameters remain almost constant over time. In other words, the following characteristic parameters of wide jams do not depend on initial conditions or on fluctuations in traffic flow:

- The flow rate out of a jam q_{out} and the vehicle density ρ_{min} in the outflow from the jam,
- The velocity of the jam’s downstream front v_g , and
- The vehicle density inside the jam ρ_{max} .

2. Jams whose widths monotonously increase over time show a tendency to the self-organization of their downstream front to the stationary moving structure described in Item 1 (9).

Figure 1 illustrates these results. The width of a jam, which spontaneously occurs near the off-ramp [Intersection 13, Figure 1(a)], monotonously increases over time during the jam’s propagation upstream [Figure 1(b)]. As a result, the jam shows the properties described earlier. In particular, in the flow-density plane the sta-

tionary movement of the downstream front of the wide jam, which occurs when free flow is formed downstream of the jam [Detector D19, Figure 1(c)], can be represented on average by line J with the coordinates (ρ_{min}, q_{out}) and $(\rho_{max}, 0)$. Line J has a slope equal to the mean value of the velocity of the jam’s downstream front v_g [Figure 1(c), line J]. It should be noted that line J , whose existence was discovered by Kerner and Rehborn (13), is not a part of the fundamental diagram: it represents the average characteristics of the downstream front of a wide jam.

When synchronized flow is formed downstream of the jam [Detector D14, Figure 1(d)], the downstream front of the jam in the flow-density plane can also be approximately represented by a line with a slope equal to velocity v_g and the right coordinates $(\rho_{max}, 0)$, nearly the same as those for line J . However, the left coordinate of this line, which is approximately related to the point $(\rho_{min}^{(syn)}, q_{out}^{(syn)})$ corresponding to the synchronized flow directly downstream of the jam, performs complex slides along line J over time (11). In the case of synchronized flow, flow rate and vehicle speed are measured immediately (with an accuracy of 1 min) after the downstream front of the jam has passed the related detector.

Metastable States of Free Traffic Flow with Respect to Jam Formation

The other result of observations of wide jams found by Kerner and Rehborn (13) is that the maximal possible flow rate in free flow $q_{max}^{(free)}$ [free flow is shown by solid points in Figure 1(c)] can be considerably higher than the flow rate out of a wide jam q_{out} [Figure 1(c)]:

$$\frac{q_{max}^{(free)}}{q_{out}} \approx 1.5 \quad (1)$$

It should be noted that the boundary (threshold) flow rate of the jam’s existence, q_b , and the corresponding threshold density, ρ_b , at which a jam either can continue to exist or be excited, respectively, are (17)

$$q_b = q_{out}, \rho_b = \rho_{min} \quad (2)$$

The mentioned properties of wide jams mean that free flow in the range of the flow rate and in the corresponding range of the vehicle density (13,17),

$$q_b \leq q \leq q_{max}^{(free)} \quad \rho_b \leq \rho \leq \rho_{max}^{(free)} \quad (3)$$

is in a metastable state. Equation 3 can be interpreted to mean that states of free flow on and above line J (i.e., states that are related to flow rates $q \geq q_b$) are metastable states of free flow and the states of free flow below line J (i.e., states that are related to flow rates $q < q_b$) are stable states with respect to the jam’s formation. To understand this conclusion, note that in the range given by Equation 3, at any given value of the vehicle density either a state of free flow or a jam may exist. Therefore, if the density is related to the range given by Equation 3, a critical local perturbation can occur in the free flow whose growth leads to a jam formation.

Recall that local perturbations with small enough amplitudes do not grow in a metastable state. However, if the amplitude of a local perturbation exceeds some critical amplitude, this critical local perturbation begins to grow. The growth of this perturbation (i.e., the nucleation effect in free flow) can lead to the formation of a jam. The flow rate q_b is the threshold of the nucleation effect: the critical

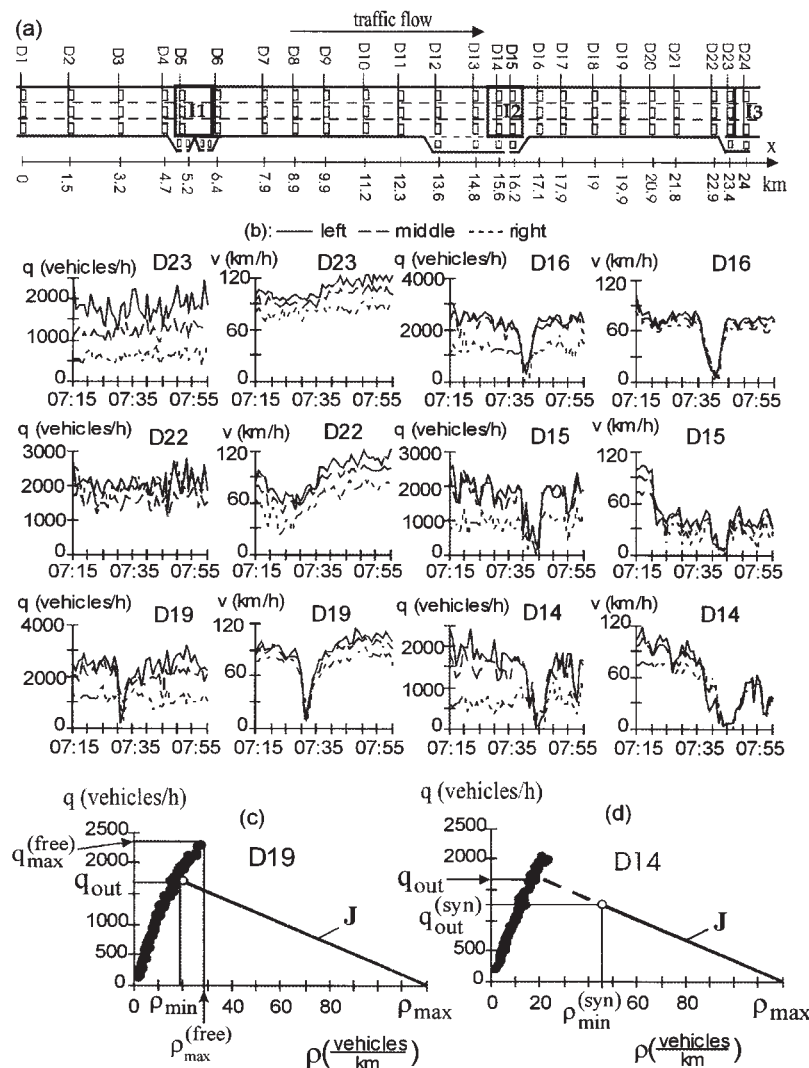


FIGURE 1 Propagation of wide jam on section of highway A5-South in Germany, schematic configuration of which is shown in (a). Data from September 13, 1995.

amplitude of the local perturbation is maximal at the threshold q_b . The higher the flow rate is, the lower the critical amplitude of the local perturbation and the higher the probability of the phase transition. Therefore, at the point $q_h = q_{\max}^{(\text{free})}$ the critical amplitude of the local perturbation is minimal, and the probability of phase transition free flow $\Rightarrow$ jam is maximal.

However, observations show (11) that independently on the flow rate, phase transition free flow $\Rightarrow$ synchronized flow (10,11) is realized considerably more frequently than phase transition free flow $\Rightarrow$ jam. Therefore, a study of the dynamic properties of synchronized flow is very important for the understanding of features of congested flow.

Dynamics of Synchronized Traffic Flow

Recently Kerner and Rehborn (9) found that synchronized flow has totally different dynamic properties in contrast to free flow and to wide jams. In particular, although the velocity of the downstream

front of a wide jam is always negative and in free flow only waves with positive velocity may propagate, in synchronized flow both waves with positive and negative velocity are possible. In addition, although the downstream front of a wide jam possesses characteristic parameters, synchronized flow has no characteristic parameters. Even the multitude of homogeneous states of synchronized flow covers a two-dimensional region in the flow-density plane: there is no fundamental diagram that may describe the whole multitude even of homogeneous states (steady states) of traffic flow. In other words, the real dynamics of synchronized flow cannot be described in the frame of the hypothesis about the fundamental diagram (9).

Properties of Homogeneous States of Free and Synchronized Flow

The correspondence of measurement points in the flow-density plane with the spatial distribution of traffic variables that has been studied

(9–11,13,14,16) [see also Figure 1(b), Figure 2(a,b)] suggests that homogeneous states of traffic flow can exist in the whole possible vehicle density range ($0 < \rho < \rho_{\max}$; $\rho_{\max}$ is the jam's density). A related example for the most interesting density range in the vicinity of the limit density $\rho_{\max}^{(\text{free})}$ of free flow is shown in Figure 2(a,b) (solid points). It can be seen that there are homogeneous-in-speed states of synchronized traffic flow, that is, states in which the average vehicle speed is nearly stationary (the speed barely changes over time), but the flow rate and consequently the density noticeably change (9) [open points in Figure 2(a)]. These states do exist on the 2.8-km-long section of the highway between Detectors D19 and D16 [Figure 2(b)]. It is important to note that states of synchronized and free flow *overlap* in densities in the vicinity of the limit density $\rho_{\max}^{(\text{free})}$ [Figure 2(a)].

Therefore, if the initial state of a flow is free and the density is increasing, then after the density exceeds $\rho_{\max}^{(\text{free})}$, the flow can nevertheless remain in a homogeneous state. In this case, however, the state of free flow has to transform into the homogeneous state of synchronized flow. Such a phase transition, free flow $\Rightarrow$ synchronized flow, occurs, for example, in the vicinity of a freeway bottleneck (10, Figure 2). This local first-order phase transition, found by Kerner and Rehborn (10), discloses the nature of the well-known “breakdown” and hysteresis phenomena, which have been studied within the last few years in connection with the problem of capacity drop at a freeway bottleneck [see references cited by Branolte (7) and Hall and Agyemang-Duah (18)]. The fundamental difference between capacity drop at a freeway bottleneck, which is in the range of 5 to 15 percent (7,18), and Equation 1, which is related to the propagation of a wide jam outside any bottlenecks, has been explained by Kerner and Rehborn (15) [see also Kerner's

explanations concerning highway capacity in the caption for his Figure 2 (11)].

Experimental data suggest that homogeneous states of synchronized flow can exist only if the amplitude of local perturbations in the flow is not very high; otherwise homogeneous states of synchronized flow can be destroyed. Note that synchronized flow itself is often destroyed in the vicinity of freeway bottlenecks. An example of such destruction of synchronized flow in the vicinity of a bottleneck (Detector D15) is shown in Figure 2(b).

Pinch Effect in Synchronized Flow

Recently reasons and features of the stop-and-go phenomenon were discovered (11): (a) First phase transition free flow $\Rightarrow$ synchronized flow occurs. (b) Then the pinch effect in synchronized flow (i.e., a self-compression of synchronized flow) is realized, in which states of flow are lying noticeably above line J in the flow-density plane and a sequence of growing narrow jams spontaneously occurs. (c) A successive process of self-transformation of the narrow jams into wide jams is realized, which determines a scale in distance between stop-and-go patterns.

On the one hand, experimental observations show that the formation of stop-and-go patterns occurs only if synchronized flow is related to states in the flow-density plane above line J (11). On the other hand, homogeneous states of synchronized flow above line J [dotted lines J in Figure 2(b)] can exist. Therefore, it may be proposed that in analogy to Equation 3, which is valid for free flow, states of synchronized flow above line J are metastable states with respect to the jam's formation.

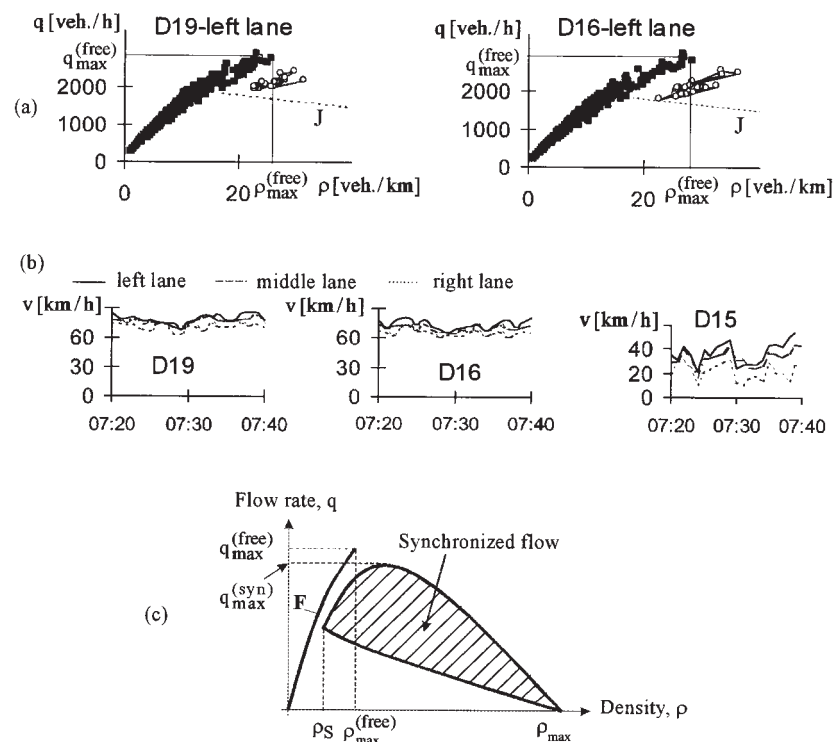


FIGURE 2 Synchronized flow: (a,b) observations from March 17, 1997; (c) hypothesis about states of free flow (curve F) and of homogeneous (steady) states of synchronized flow (hatched region).

THEORY OF CONGESTED TRAFFIC FLOW

Because of on- and off-ramps and other bottlenecks on a highway, real traffic flow is usually nonhomogeneous. However, it must be noted that all mentioned types of phase transitions in traffic flow, in particular, free flow $\Leftrightarrow$ jam, free flow $\Leftrightarrow$ synchronized flow, and synchronized flow $\Leftrightarrow$ jam(s), are also observed outside on- and off-ramps and other freeway bottlenecks (11,16). Therefore, some intrinsic nonlinear properties of homogeneous states of traffic flow, which do not depend on whether bottlenecks exist on a highway or not, are essentially responsible for the complex behavior of traffic flow. Therefore, for theoretical study of nonlinear properties of real traffic flow, the properties of homogeneous states (steady states) of traffic flow should first be understood. The hypotheses to be discussed are based on the results of observations and are devoted to these steady states of flow.

Homogeneous States of Traffic Flow

Hypothesis About Multitude of Homogeneous States (Steady States)

In the flow-density plane, homogeneous states (steady states) of flow on a multilane road are related to a curve [curve *F*, Figure 2(c)] for free flow and to a two-dimensional region [hatched region, Figure 2(c)], naturally without the jam density point $\rho_{\max}$ (and its vicinity), where the vehicle speed equals zero (i.e., no flow exists) for synchronized flow (9). The multitudes of states of free flow on a multilane road overlap homogeneous states (steady states) of synchronized flow in the density. They are separated by a gap in the flow rate at a given density (19).

Hypothesis About Behavior of Infinitesimal Perturbations in Initially Homogeneous States

Independent of the vehicle density in an initial state of flow, infinitesimal perturbations of traffic flow variables (vehicle speed, density, or both) do not grow in any homogeneous states of either free or synchronized flow or else of homogeneous-in-speed states of synchronized flow: in the whole possible range of density, homogeneous states of flow can exist. In other words, in the whole possible range of density [Figure 2(c)], there are no unstable homogeneous states of traffic flow with respect to infinitesimal perturbations of any traffic flow variables (19).

Hypothesis About Continuous Spatial-Temporal Transitions Between Different States of Synchronized Flow

Local perturbations in synchronized flow can cause continuous spatial-temporal transitions between different states of both homogeneous and homogeneous-in-speed states of synchronized flow (19) [hatched region, Figure 2(c)].

Explanation of Hypotheses for Homogeneous (Steady) States of Traffic Flow

To explain these hypotheses, note that in synchronized flow, spacing between vehicles is relatively low (i.e., the density is relatively

high) in comparison with free flow at the same flow rate [Figure 2(c)]. At low spacing a driver is able to recognize a change in the space to the vehicle in front of him even if the difference in speed is negligible. In other words, the driver is able to maintain a time-independent spacing (without taking fluctuations into account) to the vehicle in front of him in an initially homogeneous state of synchronized flow. The ability of drivers to maintain a time-independent spacing should be valid for a finite range of spacing. Therefore, a given vehicle speed may be related to an infinite multitude of homogeneous states with different densities in a limited range [for example, the range $\rho_{\min}^{(\text{syn})} \leq \rho \leq \rho_{\max}^{(\text{syn})}$ is related to homogeneous-in-speed states of synchronized flow for a fixed vehicle speed, shown by the dashed line in Figure 3(a)]. For this reason, the multitude of homogeneous states of synchronized flow covers a two-dimensional region in the flow-density plane [hatched region in Figures 2(c) and 3(a)]. States of free flow [curve *F*, Figure 3(a)] and states of synchronized flow [hatched region, Figure 3(a)] overlap in densities; that is, $\rho_{\max}^{(\text{free})} > \rho_s$, where ρ_s is the limit (minimum) density for homogeneous states of synchronized flow. However, in free flow on a multilane road, because of the possibility for vehicles to pass, the average vehicle speed can be higher than the maximum speed in synchronized flow at the same density. Therefore, there is a gap in the flow rate between states of free flow and synchronized flow at a given density. Note that states of free (curve *F*) and synchronized (hatched region) flow in Figure 3(a,c) are the same as those in Figure 2(c).

Small enough perturbations in spacing linked to an original fluctuation in the braking of a vehicle do not grow. Indeed, small enough changes in spacing are allowed; therefore, drivers should not immediately react to them. For this reason, even after a time delay, which is due to a finite reaction time of drivers τ_{react} , drivers upstream should not brake more strongly than drivers in front of them to avoid an accident. As a result, a local perturbation of traffic variables (density or vehicle speed) of small enough amplitude does not grow. An occurrence of this perturbation may cause a spatial-temporal transition to another state of synchronized flow. In other words, local perturbations may cause continuous spatial-temporal transitions between different states of synchronized flow.

Nucleation Effects and Phase Transitions in Traffic Flow

Hypothesis About Two Different Kinds of Nucleation Effects

There are two qualitatively different kinds of nucleation effects in traffic flow:

- That responsible for the jam's formation, that is, for phase transitions free flow $\Rightarrow$ jam and synchronized flow $\Rightarrow$ jam(s) [Figure 3(a,b)], and
- That responsible for phase transition free flow $\Rightarrow$ synchronized flow [Figure 3(c,d)].

The nucleation effect responsible for phase transition free flow $\Rightarrow$ synchronized flow is linked to an avalanche self-decrease in the probability of passing in traffic flow (*P*), that is, in the probability that a driver will be able to pass. This self-decrease in the probability of passing is realized if such a critical local perturbation of traffic variables (speed, density, or both) occurs that causes an initial local decrease in the probability of passing below some critical value P_{cr}

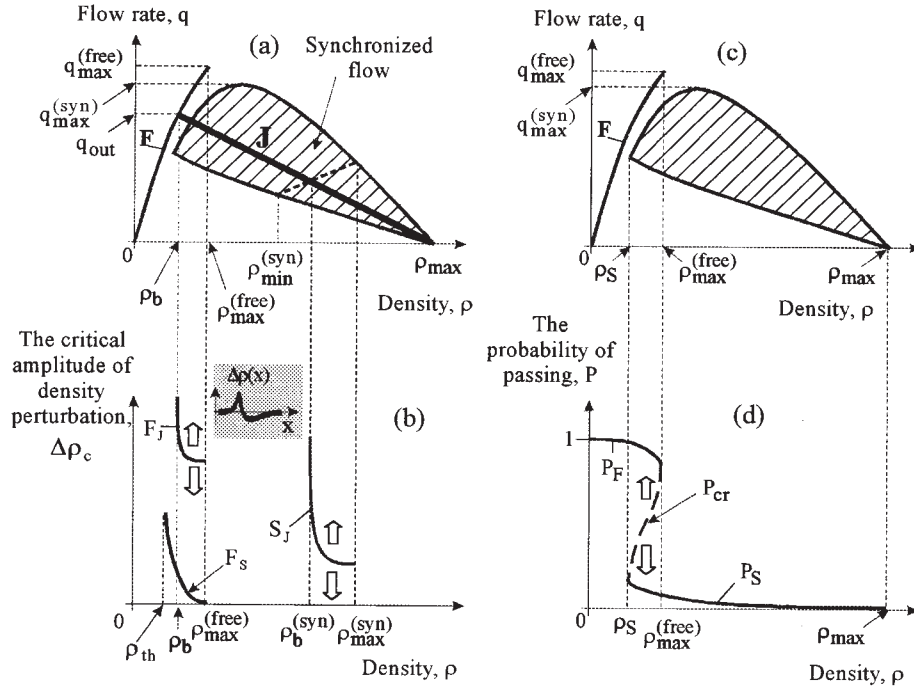


FIGURE 3 Explanations of (a, b) phase transitions free flow $\Rightarrow$ jam and synchronized flow $\Rightarrow$ jam(s) and of (c, d) phase transition free flow $\Rightarrow$ synchronized flow.

[dashed curve P_{cr} , Figure 3(d)]. The dependence of the critical amplitude of the density perturbation on the density for phase transition free flow $\Rightarrow$ synchronized flow is shown by curve F_s in Figure 3(b).

In contrast, the nucleation effect responsible for the jam's formation is linked to an avalanche self-growth of the amplitude of a local critical perturbation of the traffic variables (density, vehicle speed, or both), that is, of the local perturbation that initial amplitude exceeds some critical amplitude [dependencies of the critical amplitude of the density perturbation on the density are shown by curve F_J for phase transition free flow $\Rightarrow$ jam and by curve S_J for phase transition synchronized flow $\Rightarrow$ jam(s) in Figure 3(b)]. Curve S_J is related to a fixed vehicle speed [dashed line, Figure 3(a)].

Hypothesis About Critical Amplitude of Local Perturbations and Probability of Phase Transitions in Free Flow

At each given density in free flow, the critical amplitude of a local perturbation of traffic variables (density, vehicle speed, or both) needed for the realization of phase transition free flow $\Rightarrow$ synchronized flow [curve F_s , Figure 3(b)] is considerably lower than the critical amplitude of a local perturbation needed for the realization of phase transition free flow $\Rightarrow$ jam [curve F_J , Figure 3(a)]. Therefore, at each given density the probability of an occurrence of phase transition free flow $\Rightarrow$ synchronized flow is considerably higher than that of phase transition free flow $\Rightarrow$ jam.

The threshold density ρ_{th} for phase transition free flow $\Rightarrow$ synchronized flow can differ both from the threshold density ρ_b (Equation 3) for phase transition free flow $\Rightarrow$ jam and from the limit density ρ_s for homogeneous states of synchronized flow (Figure 3).

Because the critical value of the probability of passing is an increasing function of the density [dashed curve P_{cr} , Figure 3(d)], the higher the density is, the lower the amplitude of a local pertur-

bation of the traffic variables [density, curve F_s , Figure 3(b); vehicle speed; or both] that causes an avalanche self-decrease in the probability of passing in the related local region.

Hypothesis About Nucleation Effect Responsible for Jam Formation

Line J [Figure 3(a)] determines the threshold of the jam's existence and excitation. In other words, all (an infinite number of) homogeneous states of traffic flow related to line J in the flow-density plane are threshold states with respect to the jam's formation. Line J separates all homogeneous states of both free and synchronized flow into two qualitatively different classes (11,19):

1. In states related to points in the flow-density plane lying below line J [see axes in Figure 3(a)], no jams either can continue to exist or can be excited, and
2. States related to points in the flow-density plane lying on and above line J are metastable states with respect to the jam's formation where the related nucleation effect can be realized. Only the local perturbations of traffic variables whose amplitude exceeds some critical amplitude grow and can lead to the jam's formation [up arrows, curves F_J and S_J , Figure 3(b)]; otherwise, jams do not occur [down arrows, curves F_J and S_J , Figure 3(b)]. These critical local perturbations act as "nucleation centers" (nuclei) for the jam's formation.

The critical amplitude of the local perturbations is maximal at line J and depends both on the density and on the flow rate above line J . The critical amplitude of the local perturbations is considerably lower in synchronized flow than in free flow [compare curve F_J with curve S_J in Figure 3(b)].

The lower the vehicle speed in synchronized flow is, the lower the critical amplitude of perturbations for states of synchronized flow at the same distance above line J in the flow-density plane.

These latter two hypotheses are related to the following observations (11): (a) the “double” phase transition free flow $\Rightarrow$ synchronized flow $\Rightarrow$ jam occurs considerably more frequently than phase transition free flow $\Rightarrow$ jam and (b) jams occur considerably more frequently in the pinch region of high vehicle density.

Explanation of Hypotheses for Nucleation Effects

To explain these hypotheses, note that in free flow of low enough densities a driver is not hindered from passing; that is, the related probability of passing is $P = 1$ [Figure 3(d)]. The higher the density in free flow is, the lower the probability of passing. However, in free flow up to the limit point $\rho = \rho_{\max}^{(\text{free})}$, the probability of passing does not decrease drastically [solid curve P_F , Figure 3(d)]. Indeed, in free flow the difference between vehicle speeds on different lanes of a highway is high enough. Therefore, the probability that a vehicle will be able to pass is also high. In contrast, in synchronized flow of high enough density, drivers are not able to pass at all; that is, $P = 0$. The lower the density of synchronized flow is, the higher the probability of passing. However, in synchronized flow up to the limit point $\rho = \rho_s$, the probability of passing cannot increase drastically [solid curve P_S , Figure 3(d)]. Indeed, in contrast to free flow, in synchronized flow the difference between vehicles speeds on different lanes of a highway is low, which hinders passing.

Therefore, at each given vehicle density in the density range $[\rho_s, \rho_{\max}^{(\text{free})}]$, where states of free flow and synchronized flow overlap in the density, the probability of passing in free flow is considerably higher than that in synchronized flow. As a result, the dependence $P(\rho)$ is Z-shaped; that is, it has a hysteresis loop. Therefore $P(\rho)$ consists of three branches: (a) branch P_F for free flow, (b) branch P_S for synchronized flow, and (c) branch P_{cr} , which is related to the critical value of the probability of passing [Figure 3(d)]. The latter means that if in a local region of free flow because of a local perturbation of traffic variables (speed, density, or both) the probability of passing is decreased below the critical value P_{cr} , an avalanche self-decrease in the probability of passing occurs leading to self-formation of synchronized flow where $P = P_S$ [down arrow, Figure 3(d)]. If, on the other hand, in a local region of synchronized flow because of a local perturbation of traffic variables the probability of passing is increased above the critical value P_{cr} , an avalanche self-increase in the probability of passing occurs leading to self-formation of free flow where $P = P_F$ [up arrow, Figure 3(d)].

Because the critical value P_{cr} of the probability of passing merges with the value P_F of the probability of passing for free flow at the limit density $\rho = \rho_{\max}^{(\text{free})}$, already a local perturbation of traffic variables of a very small (but finite) amplitude causes an avalanche self-decrease in the probability of passing at the limit density $\rho = \rho_{\max}^{(\text{free})}$. Therefore, at the limit density $\rho = \rho_{\max}^{(\text{free})}$, the probability of phase transition free flow $\Rightarrow$ synchronized flow reaches 1, and the corresponding critical amplitude of a local perturbation reaches 0 [curve F_s , Figure 3(b)].

Above a threshold of any first-order phase transition, the higher the distance from the threshold is, the higher the related probability of the occurrence of the phase transition. In other words, above the related threshold density ρ_{th} [Figure 3(b)], the probability of phase transition free flow $\Rightarrow$ synchronized flow should be an increasing function of the density in free flow reaching 1 at the limit density $\rho_{\max}^{(\text{free})}$. Such

behavior of the probability of the breakdown phenomenon in a free-way bottleneck was recently discovered by Persaud et al. (20). This experimental fact confirms the conclusion by Kerner and Rehborn (10) that the latter phenomenon is caused by the first-order local phase transition free flow $\Rightarrow$ synchronized flow.

In other words, the existence of the limit point $\rho = \rho_{\max}^{(\text{free})}$ of free flow is linked to the occurrence of phase transition free flow $\Rightarrow$ synchronized flow. However, the critical amplitude of traffic variables (density, vehicle speed, or both) needed for phase transition free flow $\Rightarrow$ jam (curve F_j , Figure 3(b)) is a relatively high finite value even at the limit density $\rho = \rho_{\max}^{(\text{free})}$. Indeed, as has been mentioned, at any density the probability of phase transition free flow $\Rightarrow$ jam is considerably lower than that of phase transition free flow $\Rightarrow$ synchronized flow. To explain, recall that synchronized flow occurs if, because of a local perturbation of traffic variables (density, vehicle speed, or both), the probability of passing decreases below the critical value P_{cr} [Figure 3(d)]. The critical amplitude of a local perturbation that is able to cause the local avalanche self-decrease in the probability of passing can be considerably lower than the critical amplitude of a local perturbation needed for phase transition free flow $\Rightarrow$ jam. Indeed, for the occurrence of synchronized flow, an initial perturbation itself does not necessarily have to have an avalanche increase: it must only cause an avalanche decrease in the probability of passing. In contrast, for jam formation the amplitude of the critical perturbation itself must begin to grow an avalanche increase in the initial free flow.

An explanation of the hypothesis about the nucleation effect responsible for the jam's formation has been made by Kerner (11). Other hypotheses about features of congested flow can be found elsewhere (16,19).

CONCLUSIONS

The following conclusions resulted from this study:

- There are three different types of phases and correspondingly three different types of phase transitions in traffic flow: free flow $\Leftrightarrow$ jam, free flow $\Leftrightarrow$ synchronized flow, and synchronized flow $\Leftrightarrow$ jam(s). All of these phase transitions can be accompanied by similar-looking breakdown and hysteresis effects because these phase transitions are related to the same class of first-order phase transitions. However, they must be distinguished because nonlinear features of these phase transitions are qualitatively different. This differentiation constitutes one of the main difficulties in the understanding of traffic flow observations and theory.
- There are two different types of nucleation effects in traffic flow: that responsible for the jam's formation and that responsible for phase transition free flow $\Rightarrow$ synchronized flow.
- The nucleation effect responsible for the jam's formation is linked to an avalanche increase in the amplitude of an initial local perturbation of traffic variables (vehicle speed, density, or both). This avalanche increase occurs if the amplitude of the local perturbation exceeds some critical amplitude. At the same density the critical amplitude of the local perturbation in free flow is considerably higher than that in synchronized flow. The characteristic line of the downstream front of a wide jam determines the multitude of threshold points (an infinite number of points) of the jam's existence and the jam's excitation.
- The nucleation effect responsible for phase transition free flow $\Rightarrow$ synchronized flow is linked to an avalanche decrease in the probability of passing in free flow. This avalanche decrease

occurs if, because of an occurrence of a critical local perturbation of traffic variables, the probability of passing decreases below some critical value.

- At any density of free flow at which phase transitions may occur, the critical amplitude of the local perturbation of traffic variables (density, vehicle speed, or both) needed for phase transition free flow $\Rightarrow$ synchronized flow is considerably lower than the critical amplitude of the local perturbation of traffic variables needed for the jam's formation in free flow. This means that in free flow, phase transition free flow $\Rightarrow$ synchronized flow strongly predominates. The latter phase transition plays a decisive role in free flow, whereas phase transition free flow $\Rightarrow$ jam is relatively rare.

- In contrast to phase transition free flow $\Rightarrow$ jam, phase transition synchronized flow $\Rightarrow$ jam(s) happens often. In other words, the cascade of phase transitions free flow $\Rightarrow$ synchronized flow $\Rightarrow$ jam(s) is usually responsible for the formation of jams in an initially free flow.

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